

CELLE, Germany

WMO: 103430

Lat: 52.5958N

Lon: 10.0294E

Elev: 53

StdP: 100.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.6	-12.5	1.3	-8.9	-10.2	1.6	-6.3	11.3	8.3	10.2	8.1	3.2	90	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.8	19.4	28.7	18.6	26.6	17.8	20.4	28.0	19.5	26.6	18.6	24.9	3.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	22.0	17.0	12.2	21.7	16.1	11.5	21.1	58.4	28.2	55.6	26.6	52.8	25.0	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.9	-12.7	34.3	3.3	2.0	-15.0	35.8	-17.0	36.9	-18.8	38.1	-21.2	39.5		
(5)			-12.8	22.3	3.6	1.1	-15.4	23.1	-17.5	23.7	-19.5	24.3	-22.1	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	1.4	2.4	5.2	9.6	13.9	16.7	19.1	18.6	14.6	10.2	5.6	2.7		
	DBStd	7.32	4.74	4.47	3.70	4.01	3.84	3.50	3.41	3.23	3.14	3.61	3.76	4.62		
	HDD10.0	1109	266	214	154	57	8	0	0	0	2	42	139	227		
	HDD18.3	3169	524	447	408	263	146	74	32	37	119	253	383	484		
	CDD10.0	1121	1	1	4	44	129	200	282	266	139	48	6	2		
	CDD18.3	138	0	0	0	0	8	23	55	44	7	0	0	0		
	CDH23.3	1386	0	0	0	8	103	242	556	407	68	2	0	0		
	CDH26.7	428	0	0	0	0	19	65	198	134	13	0	0	0		
(14) Wind	WSAvg	3.0	3.6	3.5	3.5	3.0	2.9	2.8	2.8	2.6	2.6	2.8	3.0	3.4		
(15) Precipitation	PrecAvg	722	68	51	51	43	56	59	74	72	58	61	61	67		
	PrecMax	993	135	106	124	90	179	111	187	153	165	172	125	154		
	PrecMin	526	9	16	8	7	18	24	26	17	9	19	1	21		
	PrecStd	110	35	25	30	18	36	26	38	32	34	31	31	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.2	18.7	24.6	28.9	31.5	33.8	33.1	28.7	22.1	15.7	13.5		
		MCWB	10.2	9.0	11.3	14.6	17.6	20.4	20.0	19.9	18.1	16.2	13.0	10.6		
	2%	DB	10.9	11.7	15.6	21.8	26.1	28.4	31.1	30.1	25.2	19.1	13.9	12.0		
		MCWB	8.6	8.3	9.8	13.2	16.6	18.7	19.4	18.9	17.2	14.9	11.4	10.0		
	5%	DB	9.2	10.1	13.0	19.4	23.9	26.1	28.9	27.7	22.5	17.1	12.3	10.5		
		MCWB	7.4	7.4	8.7	12.1	15.1	17.5	18.8	18.4	16.1	13.7	10.1	8.6		
	10%	DB	7.8	8.8	11.1	17.1	21.6	23.9	26.4	25.2	20.3	15.6	11.0	8.9		
		MCWB	6.2	6.6	7.8	11.2	14.2	16.5	17.9	17.4	15.2	12.9	9.0	7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.2	12.0	15.5	19.4	21.3	21.5	21.2	19.3	16.7	13.3	11.3		
		MCDB	12.1	12.5	17.1	22.6	26.8	28.5	29.4	29.2	25.3	20.9	15.0	12.9		
	2%	WB	9.0	8.9	10.5	14.0	17.7	19.6	20.4	20.1	17.9	15.3	11.8	10.2		
		MCDB	10.8	10.9	14.2	19.9	24.0	26.8	28.5	27.3	23.6	18.5	13.4	11.6		
	5%	WB	7.4	7.7	9.3	12.8	16.1	18.3	19.6	19.2	16.9	14.1	10.5	8.8		
		MCDB	8.9	9.8	12.4	18.1	22.2	24.5	27.0	25.5	21.4	16.6	12.1	10.4		
	10%	WB	6.1	6.4	8.1	11.8	14.8	17.2	18.6	18.3	15.8	13.1	9.3	7.3		
		MCDB	7.5	8.4	10.8	16.5	20.1	22.4	25.1	23.9	19.7	15.3	10.8	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.2	8.0	10.1	10.7	10.3	10.3	10.4	9.4	7.7	5.3	4.7		
		MCDBR	6.0	8.2	11.9	13.8	14.0	13.9	14.3	14.3	12.9	9.7	6.5	5.4		
	5% WB	MCWBR	4.9	5.6	7.2	7.1	6.5	6.2	5.5	5.8	6.3	6.0	4.8	4.5		
		MCWBR	5.6	6.9	10.3	11.9	12.8	12.5	12.6	12.1	11.0	9.1	6.2	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.338	0.372	0.411	0.412	0.410	0.426	0.419	0.386	0.357	0.336	0.313		
		taud	2.370	2.357	2.282	2.218	2.256	2.281	2.246	2.281	2.362	2.429	2.405	2.363		
	Ebn at Noon		670	767	808	820	839	844	820	804	788	737	641	604		
		Edh at Noon	60	80	105	126	128	126	128	117	97	74	58	51		
(44) All-Sky Solar Radiation	RadAvg		0.64	1.33	2.41	3.97	4.88	5.23	4.95	4.17	3.05	1.73	0.78	0.49		
	RadStd		0.06	0.19	0.25	0.47	0.55	0.41	0.67	0.33	0.36	0.17	0.08	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (47 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	-129	N/A	N/A

Nomenclature: See separate page